Esercizio 1. *Let $C(\phi_1,..,\phi_n) \in \Delta$ boolean combination of $\phi_1(x_1,z),..,\phi_n(x_n,z)$. Then there exists $R' \in L_p$ such that $\exists y R_{C(\phi_1,..,\phi_n),\theta}(x,y) \iff R'(x)$.*

Proof. Observe that

$$a,b \models R_{C(\phi_1,..,\phi_n),\theta}(x,y) \iff a,b \models \{\neg C(\phi_1,..,\phi_n)(x,y,m) : m \in \theta(M^z)\}$$

In particular

$$\begin{aligned}
a \models \exists y R_{C(\phi_1,..,\phi_n),\theta}(x,y) &\iff b \models \{\neg C(\phi_1,..,\phi_n)(a,y,m) : m \in \theta(M^z)\} \text{ for some } b \\
&\iff \{\neg C(\phi_1,..,\phi_n)(a,y,m) : m \in \theta(M^z)\} \text{ consistent} \\
&\iff \{\neg C(\phi_1,..,\phi_n)(a,y,m) : m \in \theta(M^z)\} \text{ finitely consistent} \\
&\iff \forall k < \omega \forall m_1,..,m_k \in \theta(M^z) a \models \exists y \bigwedge_{j=1,..,k} \neg C(\phi_1,..,\phi_n)(a,y,m_j) \\
&\iff \forall k < \omega \forall m_1,..,m_k \in \theta(M^z) a \not\models \forall y \bigvee_{j=1,..,k} C(\phi_1,..,\phi_n)(a,y,m_j) \\
&\iff \forall k < \omega a \models R_{\forall y \bigvee_{j=1,..,k} C(\phi_1,..,\phi_n)(x,y,z_j), \bigwedge_{j=1,..,k} \theta(z_j)}(x) \\
&\iff a \models q(x)
\end{aligned}$$

where $q(x) := \{R_{\forall y \bigvee_{j=1,..,k} C(\phi_1,..,\phi_n)(x,y,z_j), \bigwedge_{j=1,..,k} \theta(z_j)}(x) : k < \omega\}$. Since a is arbitrary, we have $q(x) \longleftrightarrow \exists y R_{C(\phi_1,..,\phi_n),\theta}(x,y)$. By compactness, there must be $R_1,..,R_l \in q(x)$ such that $\bigwedge_{r=1,..,l} R_r(x) \longleftrightarrow \exists y R_{C(\phi_1,..,\phi_n),\theta}(x,y)$. Since the theorem has already been proved for the conjunction case, we have that there must be $R' \in L_p$ such that $\bigwedge_{r=1,..,l} R_r(x) \longleftrightarrow R'(x)$.

□